

SULIT

DP024

DP024

Physics 2

Semester II

Session 2022/2023

2 hours

DP024

Fizik 2

Semester II

Sesi 2022/2023

2 jam



KOLEJ MATRIKULASI KOLEJ MATRIKULASI PULAU PINANG
PENANG MATRICULATION COLLEGE

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **8** halaman bercetak.
This question paper consists of 8 printed pages.

SULIT

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ε_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $x = A \sin \omega t$ | 18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ |
| 2. $\omega = \frac{2\pi}{T} = 2\pi f$ | 19. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$ |
| 3. $v = A\omega \cos \omega t$ | 20. $\tau = RC$ |
| 4. $a = -A\omega^2 \sin \omega t$ | 21. $Q = Q_0 e^{\frac{-t}{RC}}$ |
| 5. $k = \frac{2\pi}{\lambda}$ | 22. $Q = Q_0 \left(1 - e^{\frac{-t}{RC}} \right)$ |
| 6. $v = f\lambda$ | 23. $B = \frac{\mu_o I}{2\pi r}$ |
| 7. $y(x, t) = A \sin(\omega t \pm kx)$ | 24. $B = \frac{\mu_o NI}{L}$ |
| 8. $v = \frac{dy}{dt}$ | 25. $B = \frac{\mu_o I}{2r}$ |
| 9. $y = 2A \cos kx \sin \omega t$ | 26. $\phi = BA \cos \theta$ |
| 10. $v = \sqrt{\frac{T}{\mu}}$ | 27. $\Phi = N\phi$ |
| 11. $I = \frac{dQ}{dt}$ | 28. $\varepsilon = -\frac{d\phi}{dt}$ |
| 12. $Q = ne$ | 29. $\varepsilon = Blv \sin \theta$ |
| 13. $\rho = \frac{RA}{l}$ | 30. $\varepsilon = -NA \frac{dB}{dt}$ |
| 14. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$ | 31. $\varepsilon = -NB \frac{dA}{dt}$ |
| 15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | 32. $R = 2f$ |
| 16. $V = IR$ | |
| 17. $C = \frac{Q}{V}$ | |

$$33. \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$34. \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$35. \quad y_m = \frac{m\lambda D}{d}$$

$$36. \quad y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$$

$$37. \quad \Delta y = \frac{\lambda D}{d}$$

1.

a) A progressive wave is described as

$$y = 2 \sin\left(4\pi\left(\frac{t}{2} + \frac{x}{0.2}\right)\right)$$

Where x and y are in cm and t is in seconds. Determine the following from this wave:

- i. Wavelength
- ii. Frequency
- iii. The speed of wave
- iv. Particle vibrational velocity when $t = 0.5$ s and $x = 3$ m

b) A stationary wave is represented by the following expression

$$y = 15 \cos 3\pi x \sin 5\pi t$$

Where y and x in centimeters and t in seconds. Determine

- i. period and wavelength.
- ii. The distance between node and antinode
- iii. the amplitude of a particle at $x = 0.5$ cm

2.

(a) A potential difference of 15 V is applied across a uniform wire of length 2.80 m and radius 0.30 cm. If 0.60 A current flows in the wire, calculate

- i. the number of free electrons flow in the wire for 2 minutes.
- ii. the resistance of the wire.
- iii. the resistivity of the wire.

(b) A battery of electromotive force (e.m.f.) 12V is connected to a network of two lamps and two resistors, as shown in **FIGURE 1**.

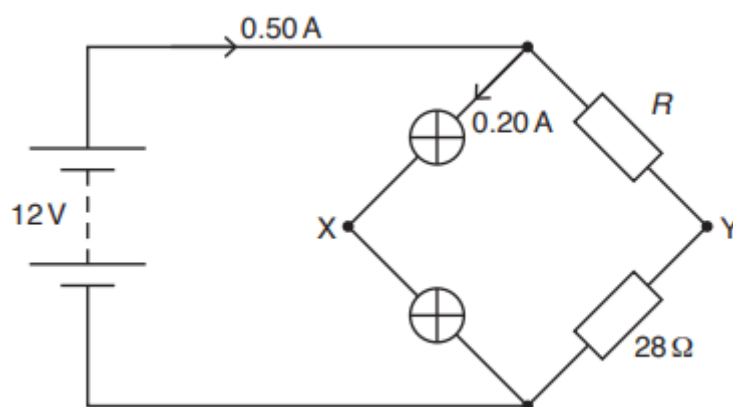
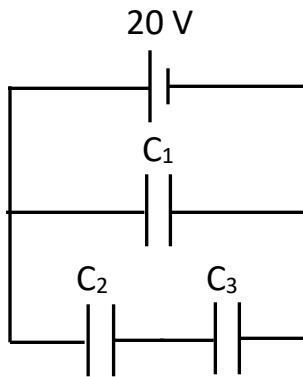


FIGURE 1

The two lamps in the circuit have equal resistances. The two resistors have resistances R and 28Ω . The lamps are connected at junction X and the resistors are connected at junction Y. The current in the battery is 0.50A and the current in the lamps is 0.20A. Determine

- i. the resistance of each lamp.
- ii. the value of resistance R .
- iii. the potential difference across 28Ω .



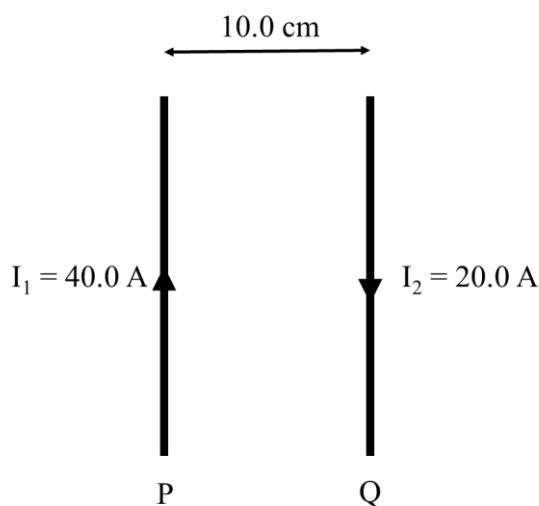
3. (a) Figure above shows a combination of three capacitors where $C_1 = 100 \mu\text{F}$, $C_2 = 22 \mu\text{F}$ and $C_3 = 47 \mu\text{F}$. A 20 V supply is connected to the combination. Determine
- (i) the effective capacitance in the circuit
 - (ii) the charge stored in the capacitor C_1
 - (iii) the potential difference across in the capacitor C_3
- (b) An uncharged capacitor of $58 \mu\text{F}$ is connected in series with a resistor of $100 \text{ k}\Omega$. Then the capacitor starts charging through the resistor. Calculate the time required for the capacitor to reach 30% of its maximum charge.

4.

(a) A solenoid length 2.0 m and 3.8 cm in diameter carries a current of 16 A. The magnetic field strength inside the solenoid is 2.3 mT.

- i. Calculate number of turns of the solenoid
- ii. Calculate the length of the wire forming the solenoid.

(b)



Two long parallel wires, P and Q are fixed on a horizontal plane in north-south direction. Wire P is 10 cm left of Q. Wire P carries 40 A current flow northward while wire Q carries 20 A current southward. Determine the magnitude and direction of the magnetic field P on wire Q.

(c)

A narrow coil with 10 turns with diameter of 4 cm is placed in uniform magnetic field of 1.20 T. After 0.25 s, the diameter of the coil is increased to 5.3 cm.

- i. Calculate the change in the area of the coil.
- ii. Calculate the maximum induced emf
- iii. If the coil has a resistance of 2.4Ω , determine the induced current in the coil
- iv. Calculate the total charge passing through the resistor in 0.25 s

5.

(a) An upright image twice the size of an object is formed by spherical mirror.

The distance between the image and the object is 30 cm.

- i. Find the focal length of the mirror.
- ii. What is the type of the spherical mirror?

(b) During a stage show, an upright image three times the size of a girl is produced by a spherical concave mirror of radius of curvature 50 cm. How far is the girl standing in front of the mirror?

(c) An object is placed 30 cm to the left of a concave lens of focal length 20 cm. What is the location of the image formed.

(d) A magnified inverted image is produced at a distance of 40 cm from an object by a convex lens. If the magnification is 4, find the focal length of this lens.

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- (a) In a Young's double-slits experiment, the slits are 0.3 mm apart and the distance of the screen to the slits is 50 cm. The slits are illuminated by an orange light of wavelength 622 nm.
- Calculate the separation between the adjacent bright fringes.
 - What will happen to the interference pattern if a light of shorter wavelength is used instead? Explain your answer.
- (b) Monochromatic light is incident on a double slit having a slit separation of 0.25 mm. an interference pattern is formed on a screen placed 2.0 m from the double slit. The distance of the first bright fringe from the central maximum is 4.6 mm.
- Find the wavelength of the light used.
 - Find the distance from the 8th bright fringe to the 3rd dark fringe on the same side of the pattern.
- (c) In Young's double-slit experiment, the light source simultaneously emits blue light of wavelength 400 nm and yellow light wavelength 600 nm. The slits are 0.08 mm apart and the interference pattern is observed on the screen at the distance of 60.0 cm from the slits. Calculate the distance between the first blue and the second yellow fringes on the same side of the pattern.